Inventiones math. 59, 205–213 (1980)

Inventiones mathematicae
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An Upper Estimation for Topological Entropy of Diffeomorphisms

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§0. Introduction

The aim of the present paper is to prove the following

Theorem. *Let f be a $C^{1+\varepsilon}$ diffeomorphism of a compact, smooth, boundaryless, Riemannian manifold M. Then*

$$h_{\text{top}}(f) \leqq \limsup_{n \to +\infty} n^{-1} \log \int_M \|Df^n(x)^\wedge\| \, d\mu(x) \qquad (*)$$

where $h_{\text{top}}(f)$ is the topological entropy of f, μ is a Riemannian measure on M induced by a given Riemannian metric, $Df^n(x)^\wedge$ is a mapping between exterior algebras of the tangent spaces $T(x)$ and $T(f^n(x))$, induced by the $Df^n(x)$. $\|\cdot\|$ is the norm on operators, induced from the Riemannian metric.

Of course the right side of (*) is independent of the Riemannian metric. This quantity does not change if $\int \|Df^{n\wedge}\|$ is replaced by $\int \max_{1 \leqq q \leqq m = \dim M} \|Df^{n\wedge q}\|$ ($^{\wedge q}$ means the q-th exterior power) or by $\max_{1 \leqq q \leqq m} \int \|Df^{n\wedge q}\|$. Geometrically $\|Df^n(x)^\wedge\|$ is the maximum of the volume of the images under $Df^n(x)$ of the arbitrarily dimensional cubes with volume 1.

The first attempt to compare the both sides of (*) was made by Sacksteder and Shub in [13]. Later, Misiurewicz found for each positive integer r a 2-sphere diffeomorphism of class C^r where the $h_{\text{top}}(f)$ is strictly smaller that the right hand side of (*).

Following Sacksteder and Shub, we shall denote the right hand side of (*) by $h_1(f)$.

Inequality (*) improves the estimation $h_{\text{top}}(f) \leqq \log \sup_{x \in M} \|Df(x)^\wedge\|$ proved by Katok [4], Ruelle [12] and Krzyżewski [5]. (However the latter inequality holds for any C^1-mapping.)

In [2] Gromov introduced an invariant

$$\text{lovl}(f) = \limsup_{n \to +\infty} n^{-1} \log \text{Vol}_m \, \text{graph}_n f$$

0020-9910/80/0059/0205/$02.00

where $\operatorname{graph}_n f=\{(x_0,\ldots,x_n)\in M\times\ldots\times M\colon x_j=f(x_{j-1}),\ j=1,\ldots,n\}$, m denotes the dimension of M and Vol_m denotes the m-dimensional volume (Hausdorff measure) in $M\times\ldots\times M$.

It is clear that for any C^1-mapping f

$$h_1(f)\leqq \operatorname{lovl}(f). \tag{**}$$

It is a straightforward to prove that $\operatorname{lovl}(f)=h_2(f)=\limsup_{n\to\infty} n^{-1}\log$ (the m-th Hausdorff measure of M in the metric d_n) where $d_n(x,y)=\max_{0\leqq i\leqq n} d(f^i(x),f^i(y))$ and d is a metric on M induced by the given Riemannian metric. The definition of $h_2(f)$ appears in [13]. So (**) follows also from $h_1(f)\leqq h_2(f)$ (see [13]).

M. Gromov proves that $h_{\mathrm{top}}(f)\leqq\operatorname{lovl}(f)$ for any holomorphic mapping f on a Kähler manifold M. So (*) together with (**) improves and extends his inequality to far larger class of mappings (but only for one-one mappings).

The Gromov's idea is that for any $x\in M$.

$$\mathrm{Vol}_m\{(y_0,\ldots,y_n)\in\operatorname{graph}_n f\colon y_0\in B_{d_n}(x,\alpha)\}>C>0$$

where C is independent of n and x and depends only on α and geometry of M. (In this paper $B_\rho(x,r)$ always denotes a ball centered at the point x of radius r in the metric ρ).

We use Gromov's idea here and prove that for some points x

$$\int_{B_{d_n}(x,\alpha)} \|Df^n(y)\wedge\|\,d\mu(y)>C\cdot\alpha^m(1-\delta)^n$$

for arbitrarily small δ.

Although this estimation is not true for all $x\in M$, it holds outside a "bad" set that is arbitrarily "small". In order to deal with these concepts one needs the ergodic theory of differentiable dynamical systems [8–11] and an estimate that shows that by excluding "bad" trajectories we neglect only arbitrarily small "portion" of entropy. Along the "good" trajectories we have partially hyperbolic behaviour and we can use Hirsch, Pugh, Shub [3] technique.

§ 1. Technical Preparation

Let f be a C^1-diffeomorphism on M, preserving a probabilistic measure ν (ν need not be equivalent to the Riemannian measure). We will assume ergodicity. Ergodic theory of differentiable dynamical systems gives the following

1.1. **Lemma.** *There exist a set $Z(\nu)$, $\nu(Z(\nu))=1$, real numbers $\lambda_1,\ldots,\lambda_s>0$, positive integers $r(1),\ldots,r(s)$ and for each $x\in Z(\nu)$ a splitting $T_xM=E_{x,1}\oplus\ldots\oplus E_{x,s}$ with $\dim E_{x,i}=r(i)$ and bases V_x^i $(i=1,\ldots,s)$ for $E_{x,i}$ respectively, such that the following statement holds. For any $\delta>0$ there exists a finite measurable function $C^1(x)\geqq 1$ such that for every $n\geqq 0$*

$$(C^1(x))^{-1}(\lambda_i-\delta)^n \leqq \|Df^n(v)\| \leqq C^1(x)\cdot(\lambda_i+\delta)^n \quad \text{when } v\in V_x^i$$

and

$$\mathrm{Vol}(Df^n(v_1),\ldots,Df^n(v_t))\cdot\|Df^n(v_1)\|^{-1}\ldots\|Df^n(v_t)\|^{-1}\geqq(C^1(x))^{-1}(1-\delta)^n$$

when $v_1,\ldots,v_t\in\bigcup_{i=1}^{s} V_x^i$ *and are pairwise different.*

1.2. **Corollary.** *For any* $\delta>0$ *there exists a finite measurable function* $C^2(x)$ *on* $Z(v)$ *so that following statement holds. For any* $x\in Z(v)$, $n\geqq 0$, *there exists an operator* $K_n(x)$: $T_{f^n(x)}M\to R^m$ *such that* $\|K_n(x)\|$, $\|K_n(x)^{-1}\|\leqq C^2(x)\cdot(1+\delta)^n$ *and the matrices of the operators* $K_{n+1}(x)\circ Df(f^n(x))\circ K_n(x)^{-1}$ *are diagonal with eigenvalues* $\lambda_1,\ldots,\lambda_s$ *having multiplicities* $r(1),\ldots,r(s)$.

The next Lemma expresses an idea similar to the λ-lemma in hyperbolic theory. The second part of our Lemma will be the key point in the proof of estimation (*).

Denote by H the set of all u-dimensional submanifolds of $R^m=R^s\times R^u$ which are graphs of C^1-functions from subsets of R^u to R^s with Lipschitz constant $\leqq 1$. Let $H(\alpha,\beta)=\{\text{graph } F\in H\colon \text{domain } F\supset B(0,\alpha),\ \sup\{\|F(x)\|\colon x\in B(0,\alpha)\}\leqq\beta\}$. Denote by p the orthogonal projection of R^m onto R^u. Denote by μ_k the k-th Lebesgue measure on R^k.

We shall denote the balls $B(0,\alpha)$ in the Euclidean spaces R^k which appear in the future ($k=s,u,m$) by $B^k(0,\alpha)$ whenever the value of k is not clear from the context.

1.3. **Lemma.** *Fix two positive numbers* λ,δ_0 *such that* $\lambda<1$, $\delta_0\leqq(1-\lambda)/5$. *For each* $n\geqq 0$ *let* A_n *be a matrix of the form* $\begin{pmatrix} A_n^s & 0\\ 0 & A_n^u\end{pmatrix}$ *in the splitting* $R^m=R^s\times R^u$, $\|A_n^s\|\leqq\lambda$ *and* $\|(A_n^u)^{-1}\|\leqq 1$. *For each* $n\geqq 0$ *let* f_n: $R^m\to R^m$ *be a diffeomorphism of class* C^1 *such that* $f_n(0)=0$ *and*

$$\rho_{C^1}(f_n,A_n)\leqq\delta_0. \tag{i}$$

Then

1° $$S\in H \quad \text{implies} \quad f_n(S)\in H \tag{1}$$

Moreover, if $S\in H(\alpha,\alpha)$, *then for every* $n\geqq 0$

$$f_n((f_{n-1}\circ\ldots\circ f_0(S))\cap p^{-1}B(0,\alpha\cdot(1-3\delta_0)^n))\in H(\alpha\cdot(1-3\delta_0)^{n+1},2\alpha). \tag{2}$$

2° *For every* $n\geqq 0$ *denote* $B_{n,\alpha}=\bigcap_{i=0}^{n}(f_{i-1}\circ\ldots\circ f_0)^{-1}B^m(0,\alpha)$. *Then*

$$\int_{B_{n,\alpha}}\|D(f_{n-1}\circ\ldots\circ f_0)(z)^{\wedge u}\|\,d\mu_m(z)\geqq C_1\cdot\alpha^m(1-3\delta_0)^{u\cdot n} \tag{3}$$

(*for* $C_1=(1/3m)^m$).

Proof. A proof of (1) is standard (see [3]), so we omit it. We prove (2) by induction:

Assume (2) holds for $n=k-1$. Then it follows that

$$S_k=f_{k-1}\circ\ldots\circ f_0(S)\cap p^{-1}B(0,\alpha\cdot(1-3\delta_0)^k)$$

intersects R^s at a unique point y_k.

Assume also that $\|y_k\|\leqq(\lambda+2\delta_0)^k\cdot\alpha$.

We have $p(S_k)\supset B(0,\alpha\cdot(1-3\delta_0)^k)$. Since $\|D(p\circ f_k\circ(p|_{S_k})^{-1})^{-1}\|\leqq(1-2\delta_0)^{-1}$ we have

$$\begin{aligned}p\circ f_k(S_k)&\supset B(0,\alpha\cdot(1-3\delta_0)^k\cdot(1-2\delta_0)-\|(A_k-f_k)(y_k)\|)\\&\supset B(0,(1-3\delta_0)^{k+1}\cdot\alpha).\end{aligned}$$

$$\|y_{k+1}\|\leqq\|A_k(y_k)\|+2\|(A_k-f_k)(y_k)\|\leqq(\lambda+2\delta_0)^{k+1}\cdot\alpha.$$

Now we prove (3).

For each $y\in B^s(0,\alpha/3)$, denote

$$S(y)=(R^u+y)\cap((f_{n-1}\circ\ldots\circ f_0)^{-1}(p^{-1}B(0,(\alpha/3)\cdot(1-3\delta_0)^n))).$$

By (2) (applied for $\alpha/3$ rather than α), for every i: $0\leqq i\leqq n$ we have

$$f_{i-1}\circ\ldots\circ f_0(S(y))\subset B^m(0,\alpha)$$

and

$$\mathrm{Vol}_u(f_{n-1}\circ\ldots\circ f_0(S(y)))\geqq(1/u)^u((\alpha/3)(1-3\delta_0)^n)^u$$

(where Vol_u denotes the u dimensional volume). Of course

$$\mathrm{Vol}_u(f_{n-1}\circ\ldots\circ f_0(S(y))\leqq\int\limits_{S(y)}\|D(f_{n-1}\circ\ldots\circ f_0)(z)^{\wedge u}\|\,d\mu_u(z).$$

So

$$\begin{aligned}&\int\limits_{B_{\alpha,n}}\|D(f_{n-1}\circ\ldots\circ f_0)(z)^{\wedge u}\|\,d\mu_m(z)\\&\quad\geqq\int\limits_{B^s(0,\alpha/3)}\Big(\int\limits_{S(y)}\|D(f_{n-1}\circ\ldots\circ f_0)(z)^{\wedge u}\|\,d\mu_u(z)\Big)\,d\mu_s(y)\\&\quad\geqq\int\limits_{B^s(0,\alpha/3)}\mathrm{Vol}_u(f_{n-1}\circ\ldots\circ f_0(S(y))\,d\mu_s(y)\\&\quad\geqq(1/3m)^m\cdot\alpha^m\cdot(1-3\delta_0)^{u\cdot n}.\end{aligned}$$

§ 2. Proof of Theorem – Step 1

We now study the dynamics along "good" trajectories.

2.1. **Lemma.** *Let f be a diffeomorphism of M of class $C^{1+\varepsilon}$. Choose any number $\delta>0$ and a probabilistic, f-invariant, ergodic measure ν on M. Then there exist a set $Z(\nu,\delta)\subset M$ and a real number $C>0$ such that $\nu(Z(\nu,\delta))\geqq1-\delta$ and such that for every $x\in Z(\nu,\delta)$, nonnegative integer n and $0<\alpha<1979$*

$$\int_{B_{d_n}(x,\alpha)} \|Df^n(y)^\wedge\| \, d\mu(y) \geqq C \cdot \alpha^m \cdot (1-\delta)^n. \tag{4}$$

Proof. The proof has been virtually done in Lemma 1.3. We need only pass from the abstract trajectory to the real trajectory on M. Here we use Hölderness of the differential of f. (The same idea proves existence of local stable manifolds, see [9–11]. By Hirsch-Pugh-Shub theory they exist in the abstract situation of Lemma 1.3. Next one passes to M using the fact that shrinking of local stable manifolds with growing n is quicker than shrinking of neighbourhoods of $f^n(x)$ where (i) is valid.)

Let $\lambda_1, \ldots, \lambda_s$ and $Z(\nu)$ be as in Lemma 1.1. Denote

$$\lambda = \max\{|\lambda_j| : |\lambda_j| < 1\}.$$

Fix an arbitrary $\delta_0 > 0$ such that

$$\delta_0 \leqq (1-\lambda)/5. \tag{ii}$$

Define δ_1 by the equality

$$1-\delta_1 = (1-3\delta_0)^{1/\varepsilon}. \tag{iii}$$

Assume also

$$(1-3\delta_0)^{2m}(1-\delta_1)^m \geqq 1-\delta. \tag{iv}$$

Corollary 1.2 gives for δ_0 a function $C^2(x)$ on $Z(\nu)$ and operators $K_n(x)$. Obviously there exists $Z = Z(\nu, \delta) \subset Z(\nu)$ such that $\nu(Z) \geqq 1-\delta$ and $C^2(x)$ is upper bounded on Z. Denote this upper bound by C_2'.

Fix $x \in Z$

Let $A_n = K_{n+1}(x) \circ Df(f^n(x)) \circ K_n(x)^{-1}$. A_n splits into contracting A_n^s and its complement A_n^u (note that A_n's are in fact independent of n). Observe that all the assumptions of Lemma 1.3 which do not involve f_n, not yet defined, are satisfied (for the couple λ, δ_0).

Let $\varphi_i \colon U_i \to M$ be charts in an atlas on M and $\xi > 0$ be a number such that for any $y \in M$ there is a number $i(y)$ such that $y \in \varphi_{i(y)} U_{i(y)}$ and $B(\varphi_{i(y)}^{-1}(y), \xi) \subset U_{i(y)}$. Let L_y denote the natural inclusion of $U_{i(y)}$ into $T_y M$ such that $L_y(\varphi_{i(y)}^{-1}(y)) = 0$. Denote by C_2'' an upper bound of the norms $\|L_y\|$ and $\|L_y^{-1}\|$ for $y \in M$. Set $C_2 = C_2' \cdot C_2''$. Let $C_3 > 0$ be a constant such that

$$\|D(\varphi_j^{-1} \circ f \circ \varphi_i)(y) - D(\varphi_j^{-1} \circ f \circ \varphi_i)(y')\| \leqq C_3 \|y - y'\|^\varepsilon$$

if $y, y' \in U_i$, $f \circ \varphi_i(y)$, $f \circ \varphi_i(y') \in \varphi_j(U_j)$. Let $l_1 \geqq 1$ denote a common Lipschitz constant for all $\varphi_j^{-1} \circ f \circ \varphi_i$.

We start now to construct f_n so as to be able to use Lemma 1.3. Let

$$\tilde{f}_n = K_{n+1}(x) \circ L_{f^{n+1}(x)} \circ (\varphi_{i(f^{n+1}(x))})^{-1} \circ f \circ \varphi_{i(f^n(x))} \circ (L_{f^n(x)})^{-1} \circ (K_n(x))^{-1}$$

for $n=0, 1, \ldots$. $\tilde{f}_n$'s make sense on the domains

$$B_n = B(0, \xi/l_1 C_2(1+\delta_0)^n)$$

for $n=0, 1, \ldots$ respectively.

Let $G: R^+ \to R^+$ be a smooth bump function such that $|dG/dt| \leqq 1$, $G|_{\langle 0,1\rangle} \equiv 1$, $G|_{\langle 3,\infty\rangle} \equiv 0$. Define $G_r: R^m \to R^+$ by $G_r(y) = G(r^{-1}\|y\|)$.

For any

$$r < \xi/3 l_1 C_2(1+\delta_0)^n \tag{5}$$

define on R^m

$$\tilde{f}_{r,n} = \begin{cases} A_n + G_r \cdot (\tilde{f}_n - A_n) & \text{inside } B_n \\ A_n & \text{outside } B_n. \end{cases}$$

We estimate for any $y \in R^m$

$$\begin{aligned} \|A_n - D\tilde{f}_{r,n}(y)\| &\leqq r^{-1} \sup\{\|A_n(z) - \tilde{f}_n(z)\| : \|z\| \leqq 3r\} \\ &\quad + \sup\{\|A_n - D\tilde{f}_n(z)\| : \|z\| \leqq 3r\} \\ &\leqq 4 \cdot \sup\{\|A_n - D\tilde{f}_n(z)\| : \|z\| \leqq 3r\} \\ &= 4 \cdot \sup\{\|D\tilde{f}_n(0) - D\tilde{f}_n(z)\| : \|z\| \leqq 3r\} \\ &\leqq 4 \cdot C_3 \cdot C_2'' \cdot \|K_{n+1}\| \cdot (3r \cdot C_2'' \cdot \|K_n^{-1}\|)^{\varepsilon} \\ &\quad 15 \cdot C_3 \cdot (C_2)^2 (1+3\delta_0)^n \cdot r^{\varepsilon}. \end{aligned}$$

Let

$$r_n = \min(\xi/4 \cdot l_1 \cdot C_2 \cdot (1+\delta_0)^n, (\delta_0/15 \cdot C_3 \cdot (C_2)^2 \cdot (1+3\delta_0)^n)^{1/\varepsilon}). \tag{6}$$

Define $f_n = \tilde{f}_{r_n,n}$. We have $\rho_{C^1}(A_n, f_n) \leqq \delta_0$. Fix a nonnegative integer N.

Let $\alpha \leqq \xi/4 l_1 C_2$ and $\alpha \leqq (\delta_0/15 \cdot C_3 \cdot (C_2)^2)^{1/\varepsilon}$.

Then by (iii) and (6)

$$f_i|_{B(0, \alpha \cdot (1-\delta_1)^N)} = \tilde{f}_i|_{B(0, \alpha \cdot (1-\delta_1)^N)} \tag{7}$$

for $i=0, \ldots, N$.

Let $C_4 = 1/l_2 \cdot C_2$ where l_2 is a common Lipschitz constant for φ_j, φ_j^{-1} when we consider the Euclidean metric on R^m and the Riemannian metric on M.

Let $C_4 \cdot \alpha \cdot (1-\delta_1)^N = \beta$. Then by Lemma 1.3

$$\int_{B_{N,\beta}} \|D(f_{N-1} \circ \ldots \circ f_0)(z)^{\wedge u}\| \, d\mu_m \geqq C_1 \cdot \beta^m (1-3\delta_0)^{u \cdot N}. \tag{8}$$

We have

$$(\varphi_{i(x)} \circ L_x^{-1} \circ (K_0(x))^{-1}(B_{N,\beta}) \subset B_{d_N}(x, \alpha).$$

Hence, in view of (7), (8) gives:

$$\begin{aligned} &\int_{B_{d_n}(x,\alpha)} \|Df^N(y)^{\wedge u}\| \, d\mu(y) \\ &\quad \geqq [(l_2^{-2} \cdot C_2^{-2} \cdot (1-\delta_0)^N)^u \cdot l_2^{-m} \cdot C_2^{-m}] \cdot C_1 \cdot C_4^m \cdot \alpha^m \cdot ((1-\delta_1)^m \cdot (1-3\delta_0)^u)^N \end{aligned}$$

(the coefficient enclosed in the square brackets is the result of the difference between Euclidean metric on R^m and the Riemannian metric on M).

In view of (iv) the proof of this Lemma is finished.

§ 3. Step 2 – Global Consideration

The ideas used here are connected with Misiurewicz's proof that $h_\nu \leqq h_{\text{top}}$ [6, Proposition 4.1] and Pesin's lower estimation of h_ν for smooth ν [10, Theorem 5.1].

We recall that $h_\nu(f)$ is a measure-theoretic entropy and is defined by $h_\nu(f) = \sup\{h_\nu(f, A)$: A is a finite Borel partition of $M\}$, $h_\nu(f, A) = \lim_{n\to\infty} n^{-1} H\left(\bigvee_{j=0}^{n} f^{-j}(A)\right)$ and $H(C) = -\sum_{a\in C} \nu(a)\cdot \log \nu(a)$.

Choose an arbitrary $\delta > 0$. Since $h_{\text{top}}(f) = \sup\{h_\nu(f)$: ν is a probabilistic, f-invariant, ergodic measure on $M\}$ (see [1]), we can fix a probabilistic, f-invariant, ergodic measure ν on M such that $h_\nu(f) \geqq h_{\text{top}}(f) - \delta$.

It is easy to prove that there exists a finite Borel partition A of M such that $h_\nu(f, A) \geqq h_{\text{top}}(f) - 2\delta$ and $\nu(\bigcup_{a\in A} \operatorname{Fr}(a)) = 0$ (locally one could cut M along m pairwise orthogonal systems of parallel $m-1$ dimensional planes of measure ν equal 0).

There exists $\delta_1 > 0$ (we assume $\delta_1 \leqq \delta$ and $\log(1-\delta_1) \geqq -\delta$) such that for any Borel subset $b \subset M$ if $\nu(b) \leqq \delta_1$ then, for the partition $A_b = \{a\backslash b$: $a \in A\} \cup \{b\}$, $h_\nu(f, A_b) \geqq h_\nu(f, A) - \delta$. (This follows from the continuity of $h_\nu(f, A)$ as a function of A, see [14, Lemma 5.8].)

Fix $Z = Z(\nu, \delta_1/2)$ and C given by Lemma 2.1. Let $\alpha > 0$ be a number such that the measure ν of the "skeleton" $\bigcup_{a\in A} \operatorname{Fr}(a)$ thickened by α is less than $\delta_1/2$. Denote this set by b_0 and the set $b_0 \cup (M\backslash Z)$ by b_1. Therefore $\nu(b_1) \leqq \delta_1$ and hence $h_\nu(f, A_{b_1}) \geqq h_{\text{top}}(f) - 3\delta$.

It is well-known that $H\left(\bigvee_{j=0}^{n} f^{-j}(A_{b_1})\right) \leqq \log \operatorname{Card}\left(\bigvee_{j=0}^{n} f^{-j}(A_{b_1})\right)$. Denote $A' = A_{b_1}\backslash\{b_1\}$. Of course

$$\operatorname{Card}\left(\bigvee_{j=0}^{n} f^{-j}(A_{b_1})\right) \leqq \Big(\sum_{\substack{0\leqq j_0<\ldots<j_s\leqq n \\ 0\leqq s\leqq n}} N(n; j_0, \ldots, j_s)\Big) + 1$$

where $N(n; j_0, \ldots, j_s) = \operatorname{Card}\left(\bigvee_{t=0}^{n} f^{-j_t}(A')\right)$.

Let $N_n = N(n; i_0, \ldots, i_p) = \max\{N(n; j_0, \ldots, j_s): j_0 < \ldots < j_s\}$. Then

$$N_n \geqq 2^{-n-1} \cdot \operatorname{Card}\left(\bigvee_{j=0}^{n} f^{-j}(A_{b_1})\right).$$

For each $a \in \bigvee_{t=0}^{p} f^{-i_t+i_0}(A')$ choose a point $x(a, n) \in a$. The points $x(a, n)$ for any fixed n are $(2\alpha, n)$-separated i.e.

$$B_{d_n}(x(a,n),\alpha)\cap B_{d_n}(x(a',n),\alpha)=\emptyset \quad \text{for } a\neq a'.$$

We sum the inequality (4) of Lemma 2.1 for $x(a,n)$ over a's and obtain

$$\int_M \|Df^n(y)^\wedge\|\,d\mu(y)\geqq C\cdot\alpha^m\cdot(1-\delta_1)^n\cdot N_n$$
$$\geqq C\cdot\alpha^m\cdot(1-\delta_1)^n\cdot 2^{-n-1}\cdot\exp H\left(\bigvee_{j=0}^{n} f^{-j}(A_{b_1})\right).$$

Hence

$$h_1(f)=\limsup_{n\to+\infty} n^{-1}\log\int_M \|Df^n(x)^\wedge\|\,d\mu(x)$$
$$\geqq h_\nu(f,A_{b_1})-\delta-\log 2\geqq h_{\text{top}}(f)-4\delta-\log 2.$$

Since δ can be choosen arbitrarily small, we obtain $h_1(f)\geqq h_{\text{top}}(f)-\log 2$.

For any $g\colon M\to M$ of class C^1 and $k\geqq 0$ $k\cdot h_1(g)=h_1(g^k)$. Indeed

$$h_1(g^k)=k\cdot\limsup_{n\to\infty}(kn)^{-1}\log\int_M\|Dg^{nk}(x)^\wedge\|\,d\mu(x)$$
$$\geqq k\cdot\max_{0\leqq k_1\leqq k-1}\limsup_{n\to\infty}((kn+k_1)/kn)\cdot(kn+k_1)^{-1}$$
$$\cdot\log\Big(\inf_{x\in M}\|Dg^{k_1}(x)^\wedge\|^{-1}\cdot\int_M\|Dg^{nk+k_1}(x)^\wedge\|\,d\mu(x)\Big)=k\cdot h_1(g).$$

The inequality in the other direction is trivial.

Thus

$$h_1(f)=n^{-1}h_1(f^n)\geqq n^{-1}(h_{\text{top}}(f^n)-\log 2)=h_{\text{top}}(f)-n^{-1}\log 2$$

for any $n>0$. Hence $h_1(f)\geqq h_{\text{top}}(f)$. This finishes the proof of Theorem.

§ 4. Final Remarks

Assume additionally that in the above situation the following holds:

Property H

$h_{\text{top}}(f)=\sup\{h_\nu(f)\colon \nu$ is a probabilistic, f-invariant, ergodic measure, $0\notin$ spectrum $(f,\nu)\}$ (spectrum (f,ν) is the set of logarithms of λ's described in Lemma 1.1).

Then, iterations of elements of A' behave "hyperpolically". (In fact using continuity of families of local stable and local unstable manifolds for $x\in Z(\nu,\zeta)$ one can and should consider rectangles instead of sets of A'.)

The following problem is analogous to the theorem proved by Misiurewicz and Szlenk [7, Theorem 4] about mappings of the unit interval:

Conjecture. Let $f\colon M\to M$ be a smooth diffeomorphism with Property H satisfied. There exist "cubes" $P_n\subset M$ such that each P_n maps under f^{k_n} like an N_n-times folded "horseshoe" and $h_{\text{top}}(f)=\lim_{n\to\infty}(k_n)^{-1}\log N_n$ (for a subsequence (k_n) of the sequence of positive integers).

If $m=2$ (i.e. for compact surfaces) and $h_{\text{top}}(f)>0$ then Property H is satisfied (because if $(\lambda_1, \lambda_2)=$spectrum (f, v) and $\lambda_1, \lambda_2 \leqq 0$ or $\lambda_1, \lambda_2 \geqq 0$, then $h_v(f) \leqq 0$ or $h_v(f)=h_v(f^{-1}) \leqq 0$, see [12]). So, if $h_v(f)>0$, then $\lambda_1<0<\lambda_2$ and local stable and local unstable manifolds $W^{s(u)}_{\text{loc}}(x)$ exist for $x \in Z(v, \delta)$.

For $x \in a \in A'$ denote $W^u(x, a)=W^u_{\text{loc}}(x) \cap a$. Since $W^u_{\text{loc}}(x)$ is one-dimensional its f^n-images intersect the sets $W^u(y, b)$ $(y \in b \in A')$ "fully" except at most two of the intersections (near the ends). This gives a strong reason for Conjecture to be true.

If $m>2$ $(u>1)$, then one should cope with proving that "most" of the intersections are "full".

Note. When I completed preparing the manuscript, I was told that A. Katok can prove the above Conjecture.

See his preprint "Lyapunov exponents, entropy and periodic orbits for diffeomorphisms" University of Maryland, May 1979, on this topic.

Also the preprint by Fried and Shub "Entropy, linearity and chain-recurrence" IHES, April 1979 has just reached Warsaw. Questions very similar to those of the present paper, connected with "entropy conjecture" have been considered there.

Krzyżewski observed that in (*) lim sup can be replaced by lim inf.

Using the same proof as (*) with slight modifications, the apparently better estimate below can be shown to hold:

$$h_{\text{top}}(f) \leqq \sup \{\limsup_{n \to +\infty} n^{-1} \log \int_M \|f^{n*}(\omega)\| \, d\mu(x): \ \omega \text{ is a smooth exterior differential form on } M\}.$$

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Received August 21, 1979/Revised January 16, 1980